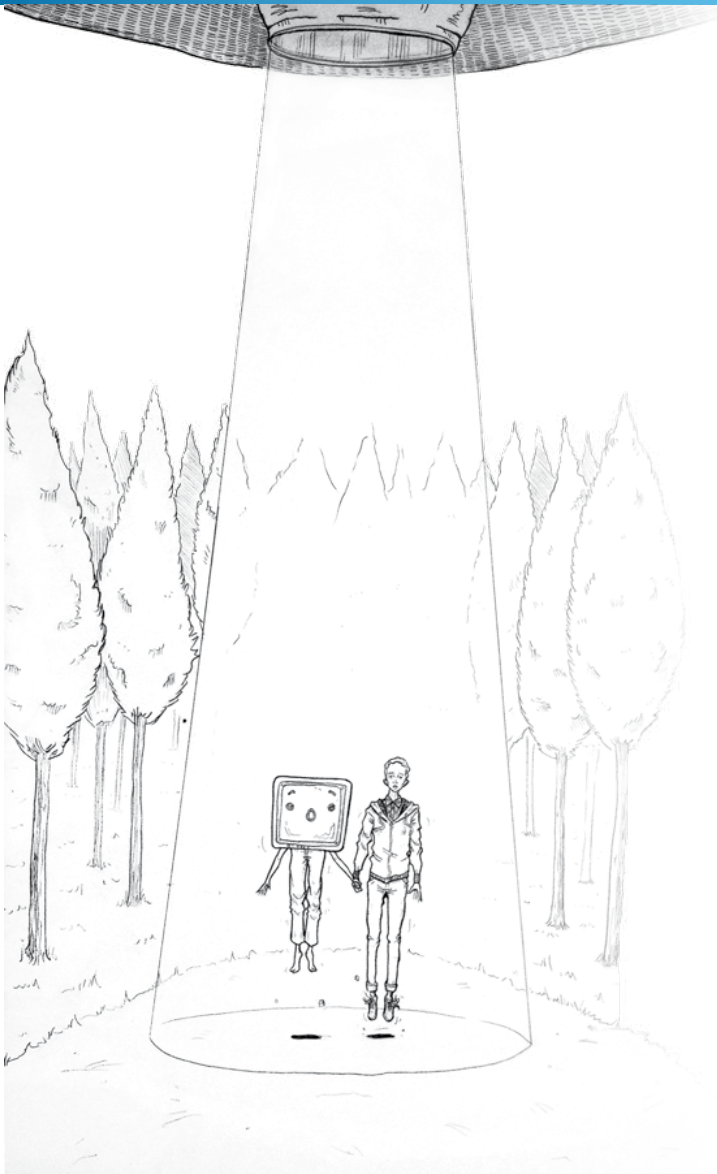




THE INTEL® PERCEPTUAL COMPUTING SDK

In October 2012 Intel introduced a comprehensive SDK that focuses on several aspects of perceptual computing, including facial recognition, voice commands, and gestures. The SDK, which includes manuals, code samples, algorithms, example applications, and tutorials, helps developers integrate perceptual computing interfaces in as simple a way as possible. Intel has always been a proponent of improved software development; in this case, they found that a combination of sensor technology and hardware computing capability created the perfect opportunity to promote perceptual computing.

The Intel Perceptual Computing SDK supports several of the most popular human-to-computer communication modalities and focuses on those that interact with a user in the 6-inch to 3-foot range of the device. While the SDK will work with many platforms, it targets Ultrabook devices, clamshell notebooks, and tablets with the use of

Perceptual Gaming

Games have pushed the technology envelope and driven innovation with user interaction. Select this category to turn a game into an experience or create an entirely new game mechanic. Enhance a game you built and love or develop something never seen before with the Intel Perceptual Computing SDK.

embedded microphones and add-on gesture recognizing cameras. (Systems with high-quality microphones and webcams with depth perception will soon be universally adopted; however, they're not yet at the miniaturized level required for small device implementation.)

Combining the efforts of Intel's internal software teams and the work of industry leaders, the Intel Perceptual Computing SDK supports a wide range of interaction types and will facilitate other software developers' many integration methods. Close-range tracking of fingers will allow developers to define usages based on a person's hand for augmented reality (grasping an object in 3D space) or for recognizing static and dynamic hand signals. Object tracking will allow developers to combine images from the camera in real-time with depth data so that "markerless" real-world objects can be used in virtual experiences. (See the "Now See This" article on augmented reality in this magazine.)

To handle face recognition and analysis, the SDK includes seven-point landmark detection and "attribution" detection including smiles, blinks, and even a user's age. Finally, the SDK's speech-recognition capabilities permit voice command and controls, as well as dictation and text-to-speech analysis.

Anil Nanduri, director of products and solutions in the Perceptual Computing Group at Intel, sees the company's development and support of the Intel Perceptual Computing SDK pushing the industry forward. "We are about driving the natural user-interaction computing capabilities and working with the ecosystem for continued advancement. We believe the SDK will drive the ecosystem in helping bring up new human-computer interface experiences."

The SDK is unique and is the first to address and combine several perceptual computing technologies, the result of Intel's collaboration with other industry leaders. Total Immersion built the computer-vision tools, Nuance provided the speech and voice components, and SoftKinetic developed the depth-tracking component. Intel combined these tools into the SDK and linked them in such a way that developers can use them easily and use them simultaneously when their applications require it. The SDK additionally includes a program for putting Creative Interactive Gesture* cameras into the hands of developers.

